

Doubling and Jamming: Universal Chern Hamiltonians on Triangulated Surfaces with Boundary

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Abstract

Higson and Prodan introduced a universal tight-binding Chern-insulator model on triangulated *closed* oriented surfaces, defined from the simplicial boundary operator B and a Hermitian Poincaré-duality operator S obtained from cap product data [1, 2]. The closed-surface Hamiltonians

$$H_{\pm} = (B + B^{\dagger}) \pm S$$

are local, self-adjoint, and carry Chern numbers ± 1 in the refinement limit.

For triangulated surfaces with boundary, the closed-case cap-product construction of S is not directly available. We treat the boundary case by a canonical *doubling-and-jamming* procedure: glue an orientation-reversed copy along $\partial\mathcal{K}$ to form the closed double $D(\mathcal{K})$, build $H_{\pm}(D(\mathcal{K}))$, and add a large on-site potential $M\mathbf{P}_{\text{copy}}$ on the copy degrees of freedom.

The unconditional part of the paper is finite-dimensional and purely linear-algebraic. We prove explicit simplex-count identities for doubling, norm-resolvent convergence with rate $O(1/M)$, a first-order Schur-complement effective operator, low-energy eigenvalue localization, and eigenvector leakage bounds. We also correct the spectral-projection statement from the earlier draft: for an isolated spectral cluster of the original-side block A , the *full* spectral projections satisfy

$$\chi_I(H_{\pm,M}) \longrightarrow \chi_I(A) \oplus 0$$

in operator norm at rate $O(1/M)$, with eventual rank stability.

For bulk–interface physics we separate what is proved here from what belongs to existing infinite-volume frameworks. Under an explicit refinement/admissibility hypothesis, doubling-and-jamming realizes a topological/trivial class-A interface whose boundary class is the image of the bulk class of the doubled universal model under the standard bulk–interface boundary map. Consequently the interface index equals the Chern jump ± 1 and changes sign under $H_+ \leftrightarrow H_-$. This is a rigorous reduction to established bulk–interface theorems rather than a new stand-alone bulk–boundary theorem for arbitrary triangulation refinements.

Numerically, we study three examples with one, two, and three boundary components, and observe bulk gaps on the doubles, in-gap states created by jamming, boundary localization, chirality reversal under $H_+ \leftrightarrow H_-$, and the predicted $1/M$ and $1/M^2$ decoupling scalings.

*Corresponding author: yitzchak.shmalo@gmail.com. For the code used in the numerical part, see the companion repository listed in the manuscript.

1 Introduction

1.1 Universal Chern Hamiltonians from simplicial data

The Haldane model is the standard prototype of a class-A (Chern) topological insulator on a regular lattice [7]. On irregular discretizations, however, direct magnetic-field or Peierls constructions often produce geometry-driven defect states that obscure clean topological gaps; see, for example, amorphous and aperiodic settings [9, 10, 11].

Higson and Prodan recently introduced a different route [1, 2]. For a triangulation of a *closed* oriented surface, one builds a local tight-binding Hamiltonian from two pieces of simplicial-topological data:

- the simplicial boundary operator B , and
- a Hermitian Poincaré-duality operator S constructed from the cap product with the fundamental cycle.

The resulting Hamiltonians

$$H_{\pm} = (B + B^{\dagger}) \pm S \quad (1.1)$$

are local and, in the refinement limit, represent nontrivial class-A phases with Chern numbers ± 1 [1, 2].

A key algebraic identity in the underlying Hilbert–Poincaré framework is

$$BS + SB^{\dagger} = 0, \quad (1.2)$$

rooted in analytic surgery theory and the work of Higson and Roe [3, 4, 5, 6]. The closed-surface model also supports the familiar topological-edge picture: if one jams or removes a region, numerics reveal interface-localized states and one-way propagation channels [1, 2].

1.2 Surfaces with boundary: why doubling-and-jamming

If the input triangulation $\mathcal{K}$ has boundary, there is no closed fundamental 2-cycle supported on the faces of $\mathcal{K}$, so the closed-case cap-product recipe for S is not immediately available. One can attempt relative constructions or boundary conditions, but the simplest route to the existing universal model is instead to pass to a closed surface by doubling.

Given a triangulated surface-with-boundary $\mathcal{K}$, we glue an orientation-reversed copy $\bar{\mathcal{K}}$ to $\mathcal{K}$ along $\partial\mathcal{K}$, thereby forming the closed double $D(\mathcal{K})$. We then build the closed universal Hamiltonians $H_{\pm}(D(\mathcal{K}))$ and add a large on-site potential on the copy side. This is the *doubling-and-jamming* construction.

1.3 What is proved here, and what is not

The paper has two distinct layers.

(i) Finite-dimensional decoupling theory. This part is unconditional. Once the doubled matrix is written in block form,

$$H_{\pm, M} = \begin{pmatrix} A & C \\ C^{\dagger} & D + M\mathbf{1} \end{pmatrix},$$

we prove explicit estimates for resolvents, spectral projections, eigenvalues, and eigenvectors. In particular, we prove a corrected low-energy spectral-subspace theorem: for isolated clusters of $\text{spec}(A)$, the *full* Riesz projections of $H_{\pm, M}$ converge to $\chi_I(A) \oplus 0$ in operator norm, and the ranks eventually agree exactly.

(ii) Asymptotic bulk–interface theory. A genuine bulk–boundary or bulk–interface statement belongs to an infinite-volume, coarse-geometric, or refinement-limit framework. We therefore isolate a precise admissibility hypothesis for a refinement family and prove that, once that hypothesis is met, doubling-and-jamming produces a topological/trivial interface whose boundary class is exactly the image of the bulk class of the doubled universal model under the standard boundary map. In other words, this paper now proves the *reduction* from surfaces-with-boundary to the established bulk–interface machinery; it does not claim a brand-new interface theorem independent of that machinery.

1.4 Contributions

For a triangulated oriented surface with boundary $\mathcal{K}$, the contributions are:

1. a canonical simplicial doubling construction and exact simplex-count identities;
2. an explicit finite-dimensional decoupling theory for jamming, with constants;
3. a corrected low-energy spectral-subspace theorem for the jammed operator;
4. an asymptotic bulk–interface reduction theorem under an explicit framework-admissibility hypothesis;
5. a reproducible numerical study on examples with one, two, and three boundary components.

2 Simplicial Hilbert space and universal operators

2.1 Triangulations and the manifold-with-boundary condition

Definition 2.1 (Triangulated surface with boundary). A finite oriented 2-dimensional simplicial complex $\mathcal{K}$ is a *triangulated surface with boundary* if every edge is incident to exactly one or two faces, and the set of edges incident to exactly one face forms a 1-dimensional simplicial subcomplex $\partial\mathcal{K}$ that is a disjoint union of cycles. Equivalently, in the graph $(\partial\mathcal{K}_0, \partial\mathcal{K}_1)$ every boundary vertex has degree exactly 2.

Remark 2.2. This is both a mathematical assumption and an implementation check. If a boundary vertex had degree different from 2, the double would develop a singularity rather than a closed 2-manifold triangulation.

2.2 Hilbert space of simplices

Let $\mathcal{K}_p$ be the set of p -simplices of $\mathcal{K}$. We work with the simplicial Hilbert space

$$\mathcal{H}(\mathcal{K}) = \ell^2(\mathcal{K}_0) \oplus \ell^2(\mathcal{K}_1) \oplus \ell^2(\mathcal{K}_2) = \bigoplus_{p=0}^2 \mathcal{H}_p(\mathcal{K}).$$

Edges are given arbitrary orientations to define the signs in the boundary operator, while faces are oriented coherently with the surface orientation. Changing these conventions only changes basis by a diagonal sign matrix and leaves the spectrum unchanged.

2.3 Boundary operator B

The simplicial boundary map B is the usual chain boundary, viewed as an operator on $\mathcal{H}(\mathcal{K})$ lowering degree by one:

$$B : \mathcal{H}_2 \rightarrow \mathcal{H}_1, \quad B : \mathcal{H}_1 \rightarrow \mathcal{H}_0, \quad B|_{\mathcal{H}_0} = 0.$$

On basis simplices,

$$\begin{aligned} B|[v_0, v_1]\rangle &= |v_1\rangle - |v_0\rangle, \\ B|[v_0, v_1, v_2]\rangle &= |[v_1, v_2]\rangle - |[v_0, v_2]\rangle + |[v_0, v_1]\rangle. \end{aligned}$$

Then $B^2 = 0$ and $B + B^\dagger$ is self-adjoint and local [1, 2].

2.4 Cap product and the Poincaré-duality operator S (closed case)

For a *closed* oriented triangulation $\mathcal{K}$ (no boundary), let

$$|X\rangle = \sum_{\tau \in \mathcal{K}_2} |\tau\rangle$$

be the fundamental 2-cycle. The cap product with $|X\rangle$ yields a degree-reversing map which, in the Higson–Prodan construction, is encoded by an operator P mapping degree p to degree $2 - p$ [1, 2]. One then defines a Hermitian operator S from P by

$$T|_{\mathcal{H}_p} = \frac{1}{2}(P^\dagger + (-1)^p P), \quad S|_{\mathcal{H}_p} = i^{p(p-1)+1} T|_{\mathcal{H}_p},$$

so that S is local, Hermitian, degree-reversing, and satisfies

$$BS + SB^\dagger = 0.$$

Remark 2.3. The relation $BS = -SB^\dagger$ does *not* by itself imply that $(B + B^\dagger)$ anticommutes with S . Any anticommutation statement used later is therefore stated explicitly as an additional hypothesis.

2.5 Universal Chern Hamiltonians (closed case)

For closed oriented $\mathcal{K}$ we define

$$H_\pm(\mathcal{K}) = (B + B^\dagger) \pm S. \tag{2.1}$$

2.6 An elementary gap criterion

The closed universal model has a refinement-stable topological gap in the Higson–Prodan theory [1, 2]. For later numerical sanity checks we record a simple algebraic implication of anticommutation.

Proposition 2.4 (Gap lower bound from invertibility of S). *Let $\mathcal{K}$ be closed. Assume S is Hermitian and satisfies*

$$(B + B^\dagger)S + S(B + B^\dagger) = 0.$$

Then

$$H_\pm(\mathcal{K})^2 = (B + B^\dagger)^2 + S^2.$$

In particular, if S is invertible then

$$\text{spec}(H_{\pm}(\mathcal{K})) \cap (-\mu, \mu) = \emptyset, \quad \mu = \frac{1}{\|S^{-1}\|}.$$

Proof. Expand $H_{\pm}^2$ and use the assumed anticommutation to eliminate the cross term. Since $(B + B^{\dagger})^2 \succeq 0$ and $S^2 \succeq \mu^2 \mathbf{1}$, we obtain $H_{\pm}^2 \succeq \mu^2 \mathbf{1}$. $\square$

3 Surfaces with boundary via doubling and jamming

3.1 Doubling

Definition 3.1 (Double). Let $\mathcal{K}$ triangulate an oriented surface with boundary. Let $\bar{\mathcal{K}}$ be a copy of $\mathcal{K}$ with the opposite surface orientation. The *double* is the simplicial complex

$$D(\mathcal{K}) = \mathcal{K} \cup_{\partial\mathcal{K}} \bar{\mathcal{K}},$$

obtained by identifying the boundary subcomplexes and keeping the faces distinct.

Theorem 3.2 (Simplex count identities under doubling). *Let $\mathcal{K}$ be a triangulated surface with boundary in the sense of Definition 2.1. If $\partial\mathcal{K}_0$ and $\partial\mathcal{K}_1$ denote the boundary vertices and boundary edges, then*

$$|D(\mathcal{K})_2| = 2|\mathcal{K}_2|, \quad |D(\mathcal{K})_1| = 2|\mathcal{K}_1| - |\partial\mathcal{K}_1|, \quad |D(\mathcal{K})_0| = 2|\mathcal{K}_0| - |\partial\mathcal{K}_0|.$$

Moreover,

$$|\partial\mathcal{K}_0| = |\partial\mathcal{K}_1|.$$

Proof. Faces are duplicated and never identified, so $|D(\mathcal{K})_2| = 2|\mathcal{K}_2|$. Interior edges and vertices are duplicated, while boundary edges and vertices are identified under gluing, yielding the formulas for $|D(\mathcal{K})_1|$ and $|D(\mathcal{K})_0|$. Since $\partial\mathcal{K}$ is a disjoint union of cycles, each connected boundary component has as many vertices as edges, and summing over components gives $|\partial\mathcal{K}_0| = |\partial\mathcal{K}_1|$. $\square$

Proposition 3.3 (Euler characteristic of the double). *Let $\mathcal{K}$ satisfy Definition 2.1. Then*

$$\chi(D(\mathcal{K})) = 2\chi(\mathcal{K}).$$

Proof. Insert the identities from Theorem 3.2 into $\chi = |0| - |1| + |2|$ and use $|\partial\mathcal{K}_0| = |\partial\mathcal{K}_1|$. $\square$

Corollary 3.4 (Genus formula for the double). *Let $\mathcal{K}$ be a connected orientable surface of genus g with $b \geq 1$ boundary components. Then the closed surface $D(\mathcal{K})$ has genus*

$$g_{\text{double}} = 2g + b - 1.$$

Proof. For an orientable surface with boundary, $\chi(\mathcal{K}) = 2 - 2g - b$. By Proposition 3.3,

$$\chi(D(\mathcal{K})) = 4 - 4g - 2b.$$

Since a closed orientable surface of genus G has Euler characteristic $2 - 2G$, one gets $G = 2g + b - 1$. $\square$

3.2 Jamming and block decomposition

On the closed double $D(\mathcal{K})$ we can construct S and hence $H_{\pm}(D(\mathcal{K}))$. We then suppress the copy degrees of freedom by a large on-site potential.

Let

$$\mathcal{H}(D(\mathcal{K})) = \mathcal{H}_{\text{orig}} \oplus \mathcal{H}_{\text{copy}},$$

where $\mathcal{H}_{\text{orig}}$ is spanned by simplices on the original side together with the glued boundary simplices, and $\mathcal{H}_{\text{copy}}$ is spanned by simplices strictly on the copy side. Let $\mathbf{P}_{\text{copy}}$ be the orthogonal projector onto $\mathcal{H}_{\text{copy}}$.

Definition 3.5 (Jammed Hamiltonian). For $M > 0$ define

$$H_{\pm, M} = H_{\pm}(D(\mathcal{K})) + M\mathbf{P}_{\text{copy}}. \quad (3.1)$$

Writing the unjammed operator in block form,

$$H_{\pm}(D(\mathcal{K})) = \begin{pmatrix} A & C \\ C^{\dagger} & D \end{pmatrix} \quad \text{on} \quad \mathcal{H}_{\text{orig}} \oplus \mathcal{H}_{\text{copy}},$$

we have

$$H_{\pm, M} = \begin{pmatrix} A & C \\ C^{\dagger} & D + M\mathbf{1} \end{pmatrix}.$$

The low-energy behaviour of $H_{\pm, M}$ is the subject of the next section.

4 Finite-dimensional decoupling theory

4.1 Schur complements

Lemma 4.1 (Schur complement resolvent formula). *Let*

$$T = \begin{pmatrix} A & C \\ C^{\dagger} & D \end{pmatrix}$$

be self-adjoint on $\mathcal{H}_1 \oplus \mathcal{H}_2$, and let $z \in \mathbb{C} \setminus \mathbb{R}$. Then $D - z$ is invertible and the $(1, 1)$ block of $(T - z)^{-1}$ is

$$(T - z)^{-1}|_{\mathcal{H}_1} = (A - z - C(D - z)^{-1}C^{\dagger})^{-1}.$$

Proof. Since D is self-adjoint and $\Im z \neq 0$, the operator $D - z$ is invertible. Standard block Gaussian elimination gives the Schur-complement formula. $\square$

4.2 Norm-resolvent convergence

Theorem 4.2 (Norm-resolvent convergence with explicit $O(1/M)$ rate). *Let $H_{\pm, M}$ be as in (3.1). For any $z \in \mathbb{C} \setminus \mathbb{R}$ and any $M > \|D\| + |\Re z|$, one has*

$$\left\| (H_{\pm, M} - z)^{-1}|_{\mathcal{H}_{\text{orig}}} - (A - z)^{-1} \right\| \leq \frac{\|C\|^2}{(M - \|D\| - |\Re z|)|\Im z|^2}. \quad (4.1)$$

In particular,

$$(H_{\pm, M} - z)^{-1}|_{\mathcal{H}_{\text{orig}}} \xrightarrow{M \rightarrow \infty} (A - z)^{-1}$$

in operator norm at rate $O(1/M)$.

Proof. By Lemma 4.1,

$$(H_{\pm, M} - z)^{-1}|_{\mathcal{H}_{\text{orig}}} = (A - z - K_M(z))^{-1}, \quad K_M(z) = C(D + M\mathbf{1} - z)^{-1}C^\dagger.$$

Because D is self-adjoint,

$$\text{spec}(D + M\mathbf{1}) \subset [M - \|D\|, M + \|D\|],$$

so

$$\|(D + M\mathbf{1} - z)^{-1}\| \leq \frac{1}{M - \|D\| - |\Re z|}.$$

Hence

$$\|K_M(z)\| \leq \frac{\|C\|^2}{M - \|D\| - |\Re z|}.$$

Now set $R_0 = (A - z)^{-1}$ and $R_M = (A - z - K_M(z))^{-1}$. The resolvent identity gives

$$R_M - R_0 = R_M K_M(z) R_0.$$

Using $\|R_0\| \leq |\Im z|^{-1}$ and $\|R_M\| \leq |\Im z|^{-1}$ yields (4.1). $\square$

4.3 A first-order effective operator

Theorem 4.3 (First-order Schur-complement expansion). *Fix $z \in \mathbb{C} \setminus \mathbb{R}$. If $M > \|D - z\|$, then*

$$(D + M\mathbf{1} - z)^{-1} = \frac{1}{M}\mathbf{1} + R_M^{(2)}(z), \quad \|R_M^{(2)}(z)\| \leq \frac{\|D - z\|}{M(M - \|D - z\|)}.$$

Consequently,

$$K_M(z) = \frac{1}{M}CC^\dagger + \tilde{R}_M(z), \quad \|\tilde{R}_M(z)\| \leq \frac{\|C\|^2 \|D - z\|}{M(M - \|D - z\|)}.$$

Defining

$$A_M^{(1)} := A - \frac{1}{M}CC^\dagger,$$

we obtain

$$\|(H_{\pm, M} - z)^{-1}|_{\mathcal{H}_{\text{orig}}} - (A_M^{(1)} - z)^{-1}\| = O(1/M^2)$$

with an explicit bound.

Proof. Write

$$D + M\mathbf{1} - z = M\left(\mathbf{1} + \frac{D - z}{M}\right).$$

If $M > \|D - z\|$, the Neumann series gives

$$\left(\mathbf{1} + \frac{D - z}{M}\right)^{-1} = \mathbf{1} - \frac{D - z}{M} + \left(\frac{D - z}{M}\right)^2 - \dots,$$

which yields the stated expansion. Multiplying on both sides by C and $C^\dagger$ gives the formula for $K_M(z)$. The final estimate follows by combining the Schur-complement formula with a resolvent identity for $A_M^{(1)} - z$. $\square$

4.4 A corrected spectral-projection theorem

The earlier draft only controlled the compression of the spectral projection to $\mathcal{H}_{\text{orig}}$, which is insufficient for rank stability of the *full* spectral projection. The next theorem fixes this.

Theorem 4.4 (Low-energy spectral-subspace convergence and rank stability). *Let $I \subset \mathbb{R}$ be compact and let Γ be a simple closed contour enclosing I and no other point of $\text{spec}(A)$. Set*

$$\delta := \text{dist}(\Gamma, \text{spec}(A)) > 0, \quad R_\Gamma := \sup_{z \in \Gamma} |\Re z|, \quad \Lambda_M := M - \|D\| - R_\Gamma.$$

Assume $\Lambda_M > 0$ and

$$\delta_M := \delta - \frac{\|C\|^2}{\Lambda_M} > 0. \quad (4.2)$$

Let

$$P_I(M) := \chi_I(H_{\pm, M}), \quad P_I(A) := \chi_I(A).$$

Then:

- (i) $\Gamma \subset \rho(H_{\pm, M})$;
- (ii) $\text{rank} P_I(M) = \text{rank} P_I(A)$;
- (iii)

$$\|P_I(M) - (P_I(A) \oplus 0)\| \leq \frac{|\Gamma|}{2\pi} \eta_M, \quad (4.3)$$

where

$$\eta_M := \frac{\|C\|^2}{\delta \delta_M \Lambda_M} + \frac{2\|C\|}{\delta_M \Lambda_M} + \frac{\|C\|^2}{\delta_M \Lambda_M^2}. \quad (4.4)$$

In particular,

$$P_I(M) \xrightarrow{M \rightarrow \infty} P_I(A) \oplus 0 \quad \text{in operator norm at rate } O(1/M),$$

and the eigenvalues of $H_{\pm, M}$ inside I converge to those of A inside I , counted with multiplicity.

Proof. For $t \in [0, 1]$ define the homotopy

$$H_{t, M} := \begin{pmatrix} A & tC \\ tC^\dagger & D + M\mathbf{1} \end{pmatrix}.$$

Fix $z \in \Gamma$. Since $D + M\mathbf{1} - z$ is invertible, the Schur complement for $H_{t, M} - z$ is

$$S_{t, M}(z) = A - z - t^2 C(D + M\mathbf{1} - z)^{-1} C^\dagger.$$

By the definition of Λ_M ,

$$\|(D + M\mathbf{1} - z)^{-1}\| \leq \frac{1}{\Lambda_M},$$

so

$$\|t^2 C(D + M\mathbf{1} - z)^{-1} C^\dagger\| \leq \frac{\|C\|^2}{\Lambda_M} < \delta.$$

Hence

$$S_{t, M}(z) = (A - z) \left(\mathbf{1} - (A - z)^{-1} t^2 C(D + M\mathbf{1} - z)^{-1} C^\dagger \right)$$

is invertible, because $\|(A - z)^{-1}\| \leq 1/\delta$. Therefore $z \in \rho(H_{t,M})$ for every $t \in [0, 1]$, and $\Gamma \subset \rho(H_{t,M})$.

The Riesz projection

$$P_I(t) := \frac{1}{2\pi i} \int_{\Gamma} (H_{t,M} - z)^{-1} dz$$

therefore depends continuously on t and has constant rank. At $t = 0$,

$$H_{0,M} = A \oplus (D + M\mathbf{1}).$$

Because Γ encloses no point of $\text{spec}(D + M\mathbf{1})$, one has

$$P_I(0) = P_I(A) \oplus 0.$$

Hence $\text{rank} P_I(M) = \text{rank} P_I(A)$, proving (ii).

For the norm estimate, abbreviate $D_M := D + M\mathbf{1}$ and

$$K_M(z) := C(D_M - z)^{-1}C^\dagger, \quad S_M(z) := A - z - K_M(z).$$

The Schur-complement inverse formula gives

$$(H_{\pm,M} - z)^{-1} = \begin{pmatrix} S_M(z)^{-1} & -S_M(z)^{-1}C(D_M - z)^{-1} \\ -(D_M - z)^{-1}C^\dagger S_M(z)^{-1} & (D_M - z)^{-1} + (D_M - z)^{-1}C^\dagger S_M(z)^{-1}C(D_M - z)^{-1} \end{pmatrix}.$$

By (4.2),

$$\|S_M(z)^{-1}\| \leq \frac{1}{\delta_M}, \quad \|(A - z)^{-1}\| \leq \frac{1}{\delta}, \quad \|(D_M - z)^{-1}\| \leq \frac{1}{\Lambda_M}.$$

Moreover,

$$S_M(z)^{-1} - (A - z)^{-1} = S_M(z)^{-1}K_M(z)(A - z)^{-1},$$

so

$$\|S_M(z)^{-1} - (A - z)^{-1}\| \leq \frac{\|C\|^2}{\delta \delta_M \Lambda_M}.$$

The off-diagonal blocks satisfy

$$\|S_M(z)^{-1}C(D_M - z)^{-1}\| \leq \frac{\|C\|}{\delta_M \Lambda_M},$$

and the extra lower-right block is bounded by

$$\left\| (D_M - z)^{-1}C^\dagger S_M(z)^{-1}C(D_M - z)^{-1} \right\| \leq \frac{\|C\|^2}{\delta_M \Lambda_M^2}.$$

Now

$$P_I(M) - (P_I(A) \oplus 0) = \frac{1}{2\pi i} \int_{\Gamma} \left[(H_{\pm,M} - z)^{-1} - \begin{pmatrix} (A - z)^{-1} & 0 \\ 0 & (D_M - z)^{-1} \end{pmatrix} \right] dz,$$

because $z \mapsto (D_M - z)^{-1}$ is analytic inside Γ and its contour integral vanishes. Estimating the block norms and using that the norm of a 2×2 block operator matrix is bounded by the sum of the norms of its blocks gives (4.3) with (4.4).

Finally, the multiset of eigenvalues inside Γ varies continuously along the homotopy $t \mapsto H_{t,M}$ while the rank of the Riesz projection remains constant, so the eigenvalues of $H_{\pm,M}$ inside I converge to those of A inside I , counted with multiplicity. $\square$

4.5 Eigenvalue localization without an isolation hypothesis

Theorem 4.5 (Low-energy eigenvalues lie near $\text{spec}(A)$). *Let E be an eigenvalue of $H_{\pm, M}$ with $|E| \leq E_{\max}$. Assume $M > \|D\| + E_{\max}$. Then*

$$\text{dist}(E, \text{spec}(A)) \leq \frac{\|C\|^2}{M - \|D\| - E_{\max}}.$$

Proof. Let $\psi = (\psi_{\text{orig}}, \psi_{\text{copy}})$ be a normalized eigenvector with eigenvalue E . Solving the copy-block equation gives

$$\psi_{\text{copy}} = -(D + M\mathbf{1} - E)^{-1}C^\dagger\psi_{\text{orig}}.$$

Substituting into the original-block equation yields

$$(A - \Sigma_M(E))\psi_{\text{orig}} = E\psi_{\text{orig}}, \quad \Sigma_M(E) = C(D + M\mathbf{1} - E)^{-1}C^\dagger.$$

Hence $E \in \text{spec}(A - \Sigma_M(E))$. Since

$$\|\Sigma_M(E)\| \leq \frac{\|C\|^2}{M - \|D\| - |E|} \leq \frac{\|C\|^2}{M - \|D\| - E_{\max}},$$

Weyl's inequality implies the claim. $\square$

4.6 Leakage into the jammed sector

Theorem 4.6 (Copy-side leakage bound). *Let (E, ψ) be an eigenpair of $H_{\pm, M}$ with $\psi = (\psi_{\text{orig}}, \psi_{\text{copy}})$ and $|E| \leq E_{\max}$. If $M > \|D\| + E_{\max}$, then*

$$\|\psi_{\text{copy}}\| \leq \frac{\|C\|}{M - \|D\| - E_{\max}} \|\psi_{\text{orig}}\|.$$

In particular, for normalized ψ ,

$$\|\psi_{\text{copy}}\| = O(1/M), \quad \|\psi_{\text{copy}}\|^2 = O(1/M^2).$$

Proof. From the copy-block equation,

$$(D + M\mathbf{1} - E)\psi_{\text{copy}} = -C^\dagger\psi_{\text{orig}}.$$

Invertibility holds because $M > \|D\| + E_{\max} \geq \|D\| + |E|$, so

$$\|\psi_{\text{copy}}\| \leq \|(D + M\mathbf{1} - E)^{-1}\| \|C\| \|\psi_{\text{orig}}\| \leq \frac{\|C\|}{M - \|D\| - E_{\max}} \|\psi_{\text{orig}}\|.$$

$\square$

Corollary 4.7 (Mean-square leakage in a window). *Fix $E_{\max} > 0$ and let $\{\psi_j\}_{j=1}^N$ be an orthonormal eigenbasis of*

$$\mathcal{E}_M := \text{Ran } \chi_{[-E_{\max}, E_{\max}]}(H_{\pm, M}), \quad N = \text{rank } \chi_{[-E_{\max}, E_{\max}]}(H_{\pm, M}).$$

If $N > 0$ and $M > \|D\| + E_{\max}$, then

$$\frac{1}{N} \sum_{j=1}^N \|\mathbf{P}_{\text{copy}} \psi_j\|^2 \leq \left(\frac{\|C\|}{M - \|D\| - E_{\max}} \right)^2.$$

Equivalently,

$$\text{tr}(\mathbf{P}_{\text{copy}} \chi_{[-E_{\max}, E_{\max}]}(H_{\pm, M})) \leq N \left(\frac{\|C\|}{M - \|D\| - E_{\max}} \right)^2.$$

Proof. Apply Theorem 4.6 to each eigenvector in the window and average the squared estimates. $\square$

Remark 4.8. Theorem 4.4 now gives a genuine operator-norm control of an isolated low-energy spectral subspace, while Corollary 4.7 is the window-averaged estimate used directly in the numerics.

4.7 Gap stability under bounded perturbations

Proposition 4.9 (Weyl-type gap stability). *Let H be Hermitian and suppose $\text{spec}(H) \cap (-g, g) = \emptyset$ for some $g > 0$. Let V be Hermitian with $\|V\| \leq \varepsilon < g$. Then*

$$\text{spec}(H + V) \cap (-(g - \varepsilon), g - \varepsilon) = \emptyset.$$

Proof. For Hermitian matrices, the Hausdorff distance between the spectra of H and $H + V$ is bounded by $\|V\|$. $\square$

5 Asymptotic bulk–interface theorem

The finite-dimensional results above show that jamming suppresses the copy sector and controls low-energy states. To turn this into a rigorous bulk–interface statement, one must place the refinement family inside a standard infinite-volume or coarse-geometric framework. We make that dependence explicit.

Assumption 5.1 (Framework admissibility). Let $(\mathcal{K}_n)$ be a refinement sequence of a fixed compact oriented surface with boundary. After the standard rescaling used to keep simplex diameters uniformly comparable, assume that:

- (a) the doubled Hamiltonians $H_{\pm}(D(\mathcal{K}_n))$ form a bounded-geometry local family with a uniform Fermi gap at 0;
- (b) their Fermi projections define a bulk class

$$[P_{\pm}] \in K_0(\mathcal{A}_{\text{bulk}})$$

in some standard class-A bulk algebra $\mathcal{A}_{\text{bulk}}$ (for example, a Roe-algebra or crossed-product/Toeplitz framework);

- (c) the partition induced by the doubled boundary determines an interface algebra $\mathcal{A}_{\text{int}}$ and boundary map

$$\partial : K_0(\mathcal{A}_{\text{bulk}}) \rightarrow K_1(\mathcal{A}_{\text{int}})$$

for which the corresponding standard bulk–interface theorem applies [12, 13];

- (d) the bulk class has Chern number

$$\text{Ch}([P_{\pm}]) = \pm 1,$$

as established for the closed universal model under refinement [1, 2].

Remark 5.2. The point of Assumption 5.1 is not to assume the conclusion for the jammed operator. Rather, it isolates the standard external hypotheses required to place the refinement family inside an existing bulk–interface theory. Once this is granted, the reduction from doubling-and-jamming to a topological/trivial interface is proved below.

Proposition 5.3 (The jammed side is topologically trivial). *Fix the Fermi level at 0. Suppose the copy-side bulk operator D_{bulk} satisfies a uniform bound $\|D_{\text{bulk}}\| \leq B$. If $M > B$, then*

$$\chi_{(-\infty, 0)}(D_{\text{bulk}} + M\mathbf{1}) = 0.$$

Hence the jammed side defines the trivial class in K_0 .

Proof. All spectrum of $D_{\text{bulk}} + M\mathbf{1}$ lies in $(0, \infty)$ if $M > B$, so the Fermi projection below 0 vanishes identically. $\square$

Theorem 5.4 (Bulk–interface reduction for doubling and jamming). *Assume Assumption 5.1. Choose M large enough that the jammed copy-side bulk is positive and therefore trivial at Fermi level 0. Then the doubled-and-jammed family defines the same interface class as a topological/trivial class-A interface, and its boundary class is*

$$\partial([P_{\pm}]) \in K_1(\mathcal{A}_{\text{int}}).$$

Equivalently, the interface index equals the bulk Chern jump,

$$\text{Index}_{\text{int}}(H_{\pm, M}) = \text{Ch}([P_{\pm}]) - 0 = \pm 1.$$

In particular, the interface class is nontrivial, and it changes sign under $H_+ \leftrightarrow H_-$.

Proof. Away from a bounded collar of the glued boundary, the jammed Hamiltonian agrees on the original side with the closed bulk phase and on the copy side with the positively shifted Hamiltonian $D_{\text{bulk}} + M\mathbf{1}$. By Proposition 5.3, the latter has zero Fermi projection and hence represents the trivial class in K_0 .

Therefore the doubled-and-jammed model is, at the level relevant to the interface theorem, a class-A interface between the bulk class $[P_{\pm}]$ and the zero class. The standard bulk–interface theorem in the chosen framework [12, 13] identifies the interface class with the boundary-map image of the bulk class difference:

$$\partial([P_{\pm}] - 0) = \partial([P_{\pm}]).$$

Assumption 5.1(d) gives $\text{Ch}([P_{\pm}]) = \pm 1$, so the interface index is nonzero and has sign ± 1 . Replacing H_+ by H_- flips the bulk Chern number and hence the sign of the interface class. $\square$

Corollary 5.5 (Componentwise edge index when a local interface formula is available). *In addition to Assumption 5.1, suppose the chosen interface framework provides a local conductance formula for curved interfaces in terms of a test window intersecting the interface, such as an intersection-number formula [15, 14]. Then any window intersecting exactly one boundary component once and no other component has local edge index ± 1 . Consequently, each boundary component supports one chiral channel with respect to its induced boundary orientation.*

Proof. Choose a measurement window meeting a single boundary component transversely once. The local formula identifies the edge index with the bulk Chern jump multiplied by the corresponding intersection number, which is 1 by construction. The sign is inherited from the choice of $H_{\pm}$ and the induced boundary orientation. $\square$

Remark 5.6. Theorem 5.4 is stronger than the earlier draft’s informal conditional paragraph: it no longer assumes a bulk–boundary statement specifically for the jammed operator. The only external input is the standard admissibility of the refinement family within an established interface framework.

Table 1: Mesh sizes for the three boundary triangulations $\mathcal{K}$ and their doubles $D(\mathcal{K})$, including boundary counts. This table is a primary correctness check for Theorem 3.2.

ex	$\#\partial$	$ \partial\mathcal{K}_0 $	$ \partial\mathcal{K}_1 $	$ \mathcal{K}_0 $	$ \mathcal{K}_1 $	$ \mathcal{K}_2 $	$\dim \mathcal{H}(\mathcal{K})$	$ D_0 $	$ D_1 $	$ D_2 $	$\dim \mathcal{H}(D)$	g
ex1	1	22	22	73	194	122	389	124	366	244	734	0.5666
ex2	2	88	88	272	728	456	1456	456	1368	912	2736	0.3380
ex3	3	78	78	329	912	582	1823	580	1746	1164	3490	0.3272

6 Numerical experiments

This section summarizes the computational pipeline and the finite-size diagnostics. The numerics are not used to prove the theorems above; they illustrate the theory and provide implementation checks.

6.1 Pipeline and reproducibility

The companion script:

1. generates boundary triangulations from a subdivided icosphere and checks Definition 2.1;
2. constructs the double and verifies Theorem 3.2 and Proposition 3.3;
3. assembles B , the cap-product operator, S , and the closed Hamiltonians $H_{\pm}(D(\mathcal{K}))$;
4. computes the bulk half-gap

$$g := \min\{|E| : E \in \text{spec}(H_+(D(\mathcal{K})))\};$$

5. forms the jammed Hamiltonians $H_{\pm,M}$, diagonalizes them, and evaluates localization, transport, and decoupling diagnostics;
6. optionally adds weak onsite disorder and repeats the transport diagnostics.

The script writes the summary tables included below when the auxiliary files are present.

6.2 Three example families

We consider three examples whose main distinguishing feature is the number b of boundary components:

1. **Example 1:** Disk-like patch (1 boundary) ($b = 1$),
2. **Example 2:** Annulus-like band (2 boundaries) ($b = 2$),
3. **Example 3:** Three-boundary patch (disk with two holes) ($b = 3$).

By Corollary 3.4, their doubles have genera 0, 1, and 2, respectively.

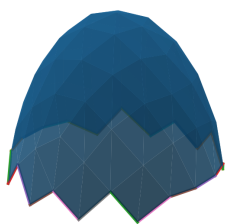
6.3 Geometry and bulk gaps

Figure 1 shows the input meshes and highlights the boundary cycles. Figure 2 shows the sorted spectra of the closed doubled Hamiltonians $H_+(D(\mathcal{K}))$ together with the bulk half-gap g . In the displayed runs all three examples are gapped.

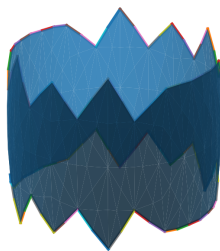
Table 2: Selected numerical outputs: the bulk half-gap g for $H_+(D(\mathcal{K}))$, the number of jammed eigenvalues selected by the finite-size criterion $|E| < g$, and fitted drift slopes when reported.

ex	g	# in-gap ($ E < g$)	v_+	v_-
ex1	0.5666	5	-0.890	0.893
ex2	0.3380	14	—	—
ex3	0.3272	12	—	—

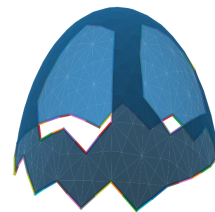
Example 1 mesh (cap, 1 boundary) [bary=0]



Example 2 mesh (band, 2 boundaries) [bary=1]



Example 3 mesh (cap+2 holes, r=0.225) [bary=1]



(a) Disk-like patch (1 boundary). (b) Annulus-like band (2 boundaries). (c) Three-boundary patch (disk with two holes).

Figure 1: Boundary triangulations used in the experiments. Boundary edges are highlighted.

6.4 Jamming, in-gap states, and localization

We form

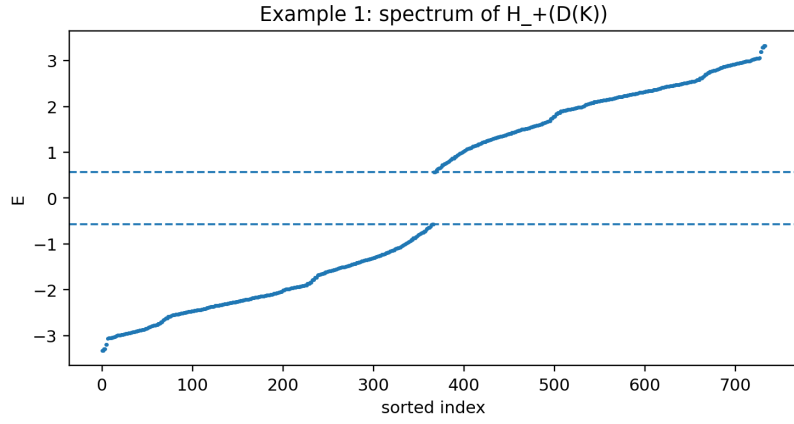
$$H_{+,M} = H_+(D(\mathcal{K})) + M\mathbf{P}_{\text{copy}}.$$

The jammed spectra are shown in Figure 3. A high-energy cluster near $E \approx M$ corresponds to copy-supported states, while additional discrete levels appear inside the inherited bulk gap window $|E| < g$. These in-gap levels are the finite-size candidates for interface states.

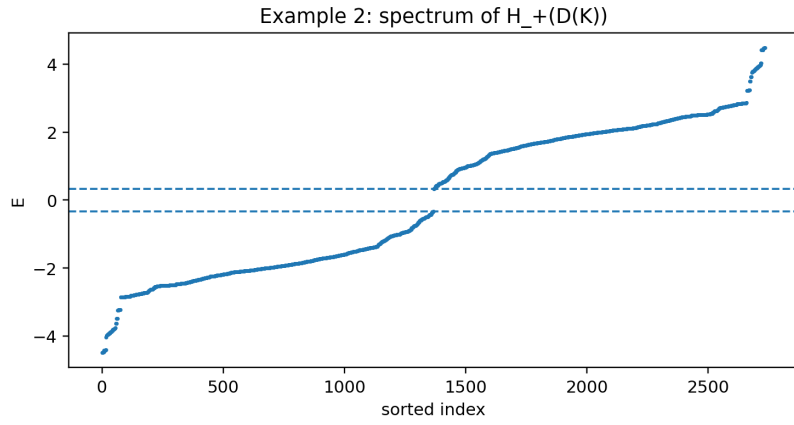
For Example 1, Figure 4 plots a boundary-weight diagnostic

$$w_{\partial}(\psi) = \|\mathbf{P}_{\partial}\psi\|^2,$$

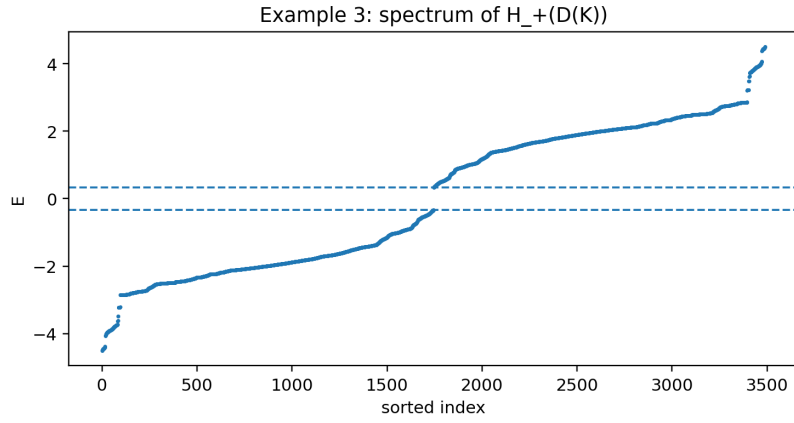
where $\mathbf{P}_{\partial}$ projects to the boundary vertices on the original side. The in-gap states carry noticeably larger boundary weight than typical bulk states, consistent with interface localization.



(a) Example 1.

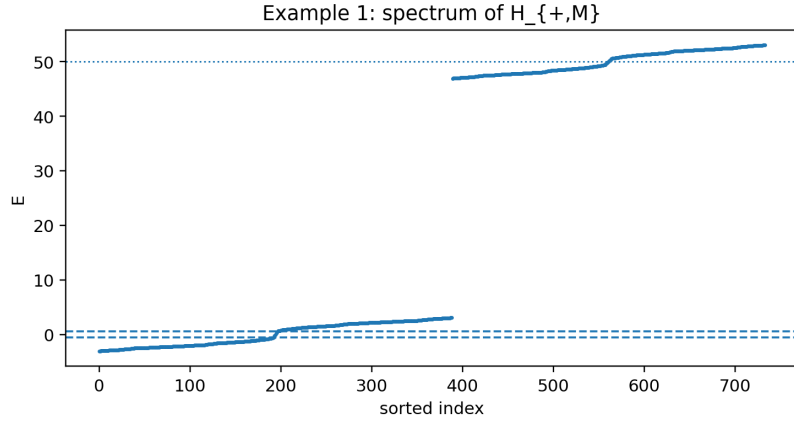


(b) Example 2.

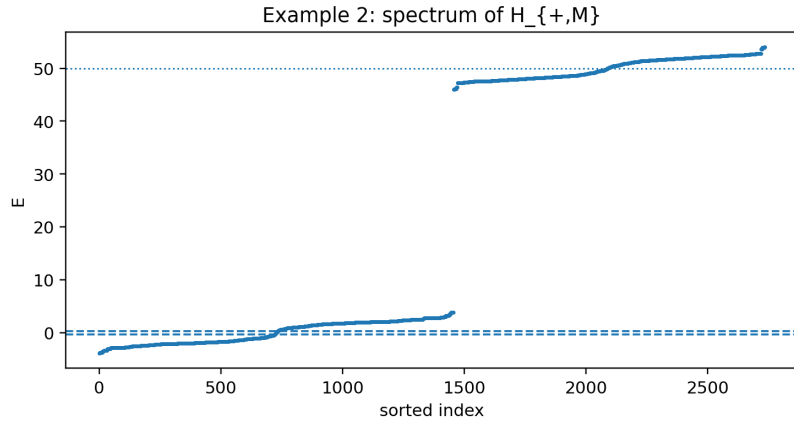


(c) Example 3.

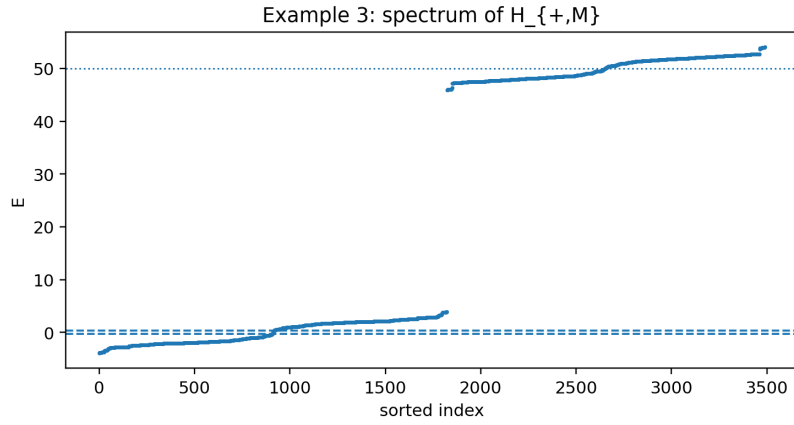
Figure 2: Sorted spectra of $H_+(D(K))$ on the doubles. Dashed lines mark $\pm g$, where $g = \min |\text{spec}(H_+(D(K)))|$.



(a) Example 1.



(b) Example 2.



(c) Example 3.

Figure 3: Spectra of the jammed Hamiltonians $H_{+,M}$. Dashed lines indicate the bulk gap scale inherited from the double.

6.5 Boundary transport and chirality reversal

To probe chirality we evolve a wavepacket,

$$\psi(t) = e^{-itH_{\pm,M}}\psi(0),$$

and project it to the in-gap subspace selected by $|E| < g$. For each boundary component, the script orders the boundary vertices cyclically and plots the boundary probability as a function of time. Diagonal ridges in the resulting heatmaps indicate unidirectional propagation.

Figure 5 displays these heatmaps for the three examples. For Example 1, Figure 6 additionally extracts a signed drift slope from the circular mean position. The sign flips under $H_+ \leftrightarrow H_-$, matching the sign flip predicted by Theorem 5.4.

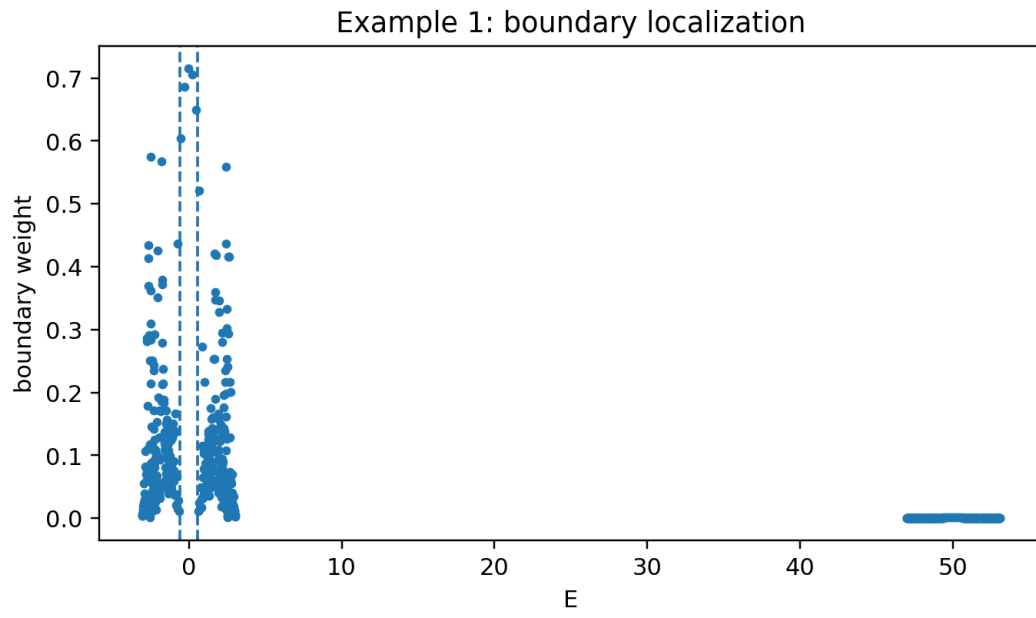


Figure 4: Example 1: boundary-weight diagnostic w_{∂} versus eigenvalue. States selected by the finite-size gap criterion $|E| < g$ show enhanced boundary weight.

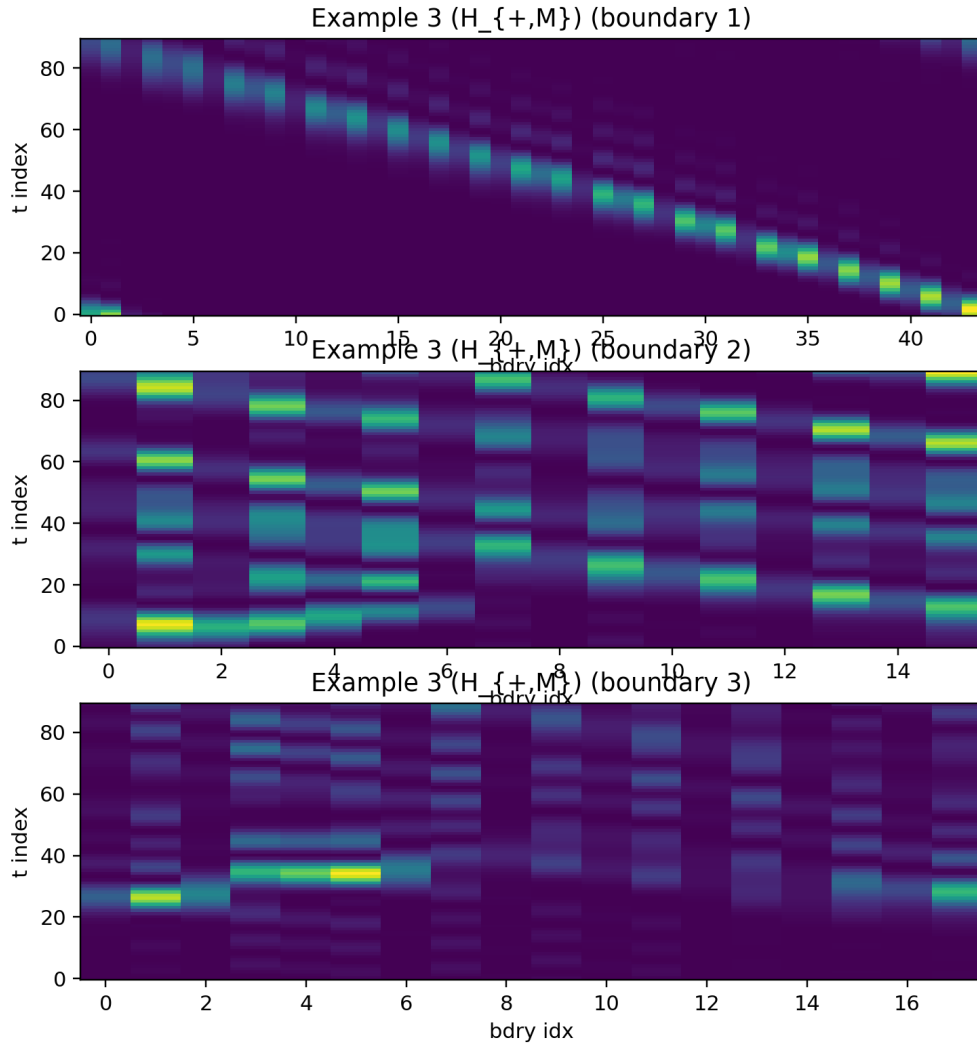
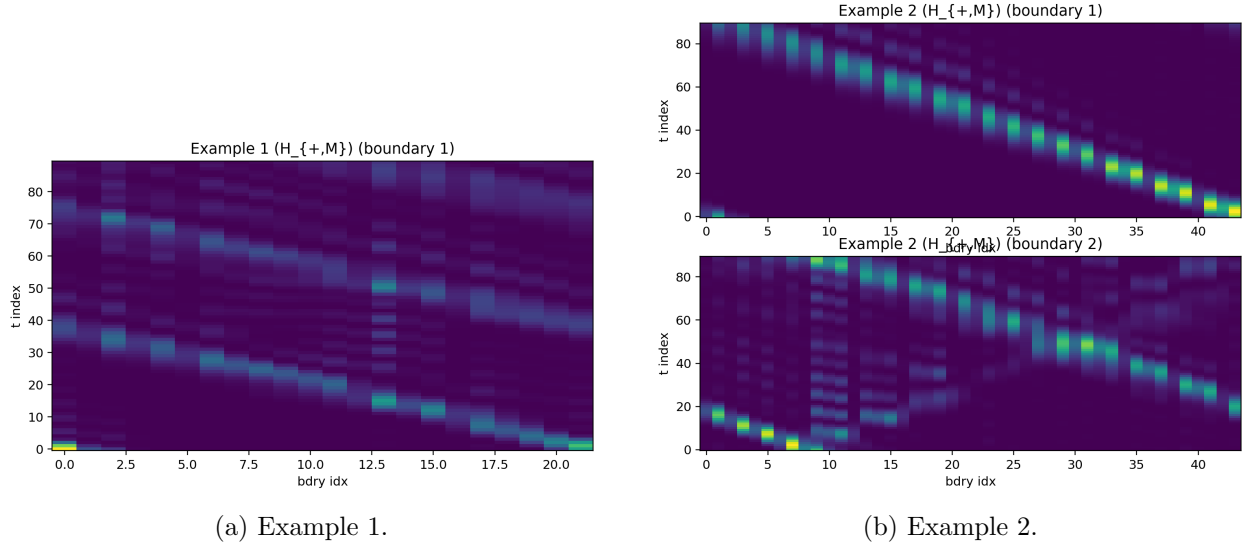


Figure 5: Boundary transport heatmaps for wavepackets projected to the finite-size in-gap subspace. Diagonal ridges indicate one-way motion along the boundary cycles.

6.6 Direct numerical checks of the decoupling theory

The analysis of Section 4 makes quantitative predictions in $1/M$.

Resolvent convergence. For a fixed non-real spectral parameter, typically $z = i$, we compute

$$\Delta(M) = \left\| (H_{+,M} - i)^{-1}|_{\mathcal{H}_{\text{orig}}} - (A - i)^{-1} \right\|.$$

Theorem 4.2 predicts $\Delta(M) = O(1/M)$.

Leakage. Theorem 4.6 and Corollary 4.7 predict that the mean squared copy weight of bounded-energy states should scale like $O(1/M^2)$.

Eigenvalue localization. Theorem 4.5 predicts an $O(1/M)$ bound on the distance of bounded-energy eigenvalues of $H_{+,M}$ from $\text{spec}(A)$.

The corresponding plots are shown in Figures 7 to 9, and the Example 1 summary table appears in Table 3. The observed trends are consistent with the theoretical predictions, and the explicit bounds are conservative but valid over the tested range.

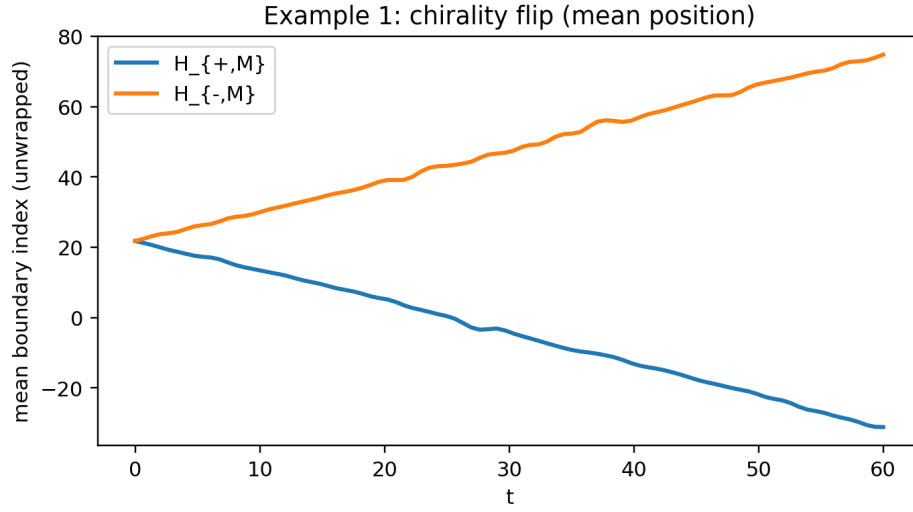


Figure 6: Example 1: opposite drift slopes for $H_{+,M}$ and $H_{-,M}$, showing chirality reversal.

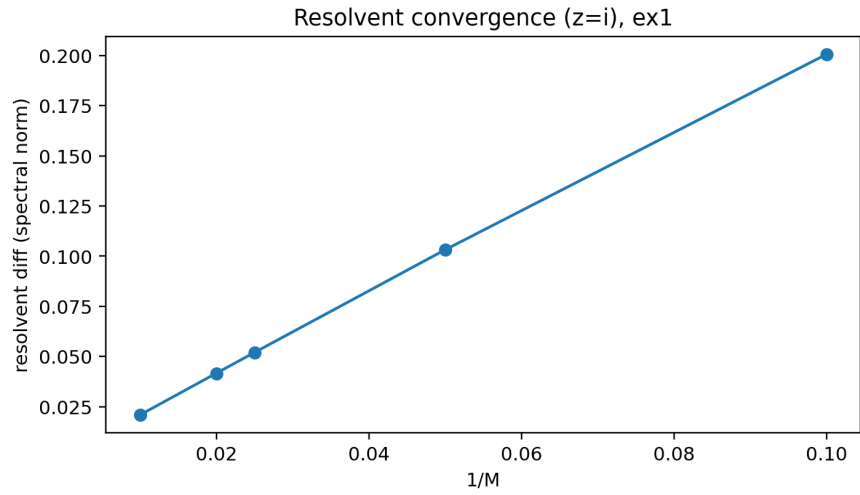


Figure 7: Example 1: empirical norm-resolvent convergence versus $1/M$, consistent with Theorem 4.2.

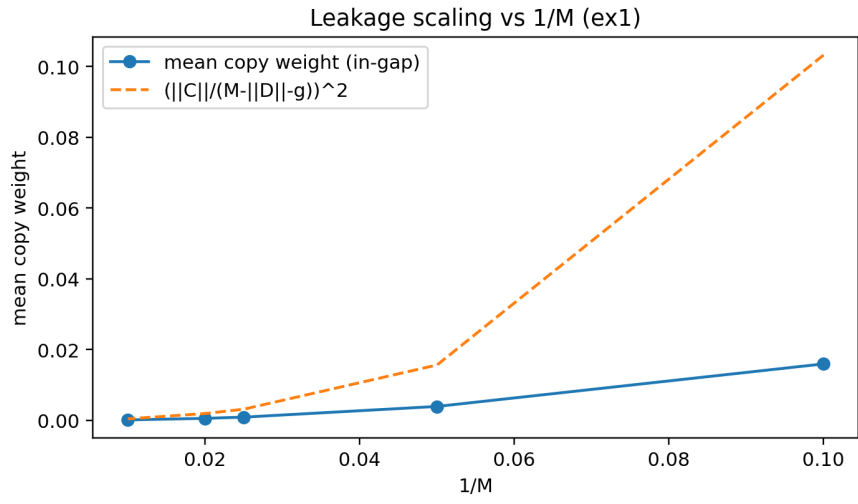


Figure 8: Example 1: mean squared copy leakage versus $1/M$. The observed scaling is consistent with Theorem 4.6 and Corollary 4.7.

6.7 Refinement and disorder

Figure 10 reports how the bulk half-gap g varies with subdivision level. This is useful because coarse meshes can have small or nearly closed gaps, whereas a stabilizing trend under refinement is evidence for a genuine bulk gap.

Figure 11 adds weak onsite disorder to the Example 1 jammed Hamiltonian. Proposition 4.9 explains why a pre-existing gap cannot close under a sufficiently small bounded perturbation. Numerically, the transport ridge persists under weak disorder, as expected for a topological edge response.

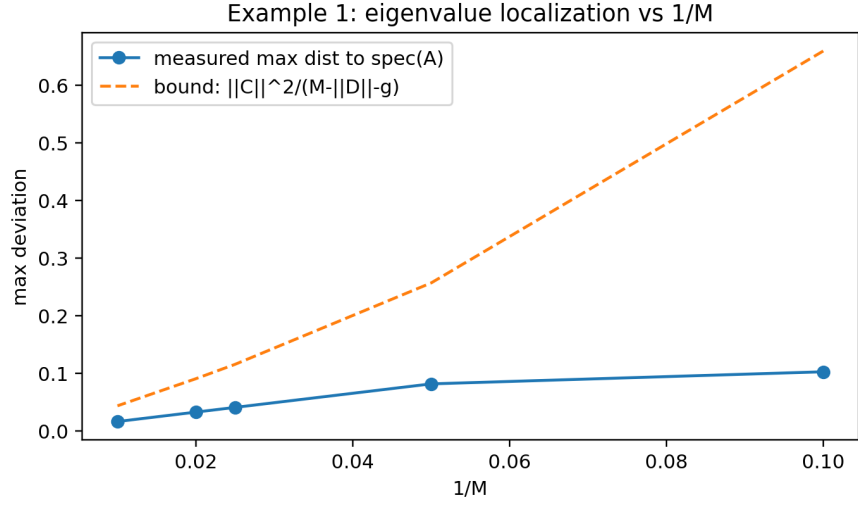


Figure 9: Example 1: measured eigenvalue localization against $1/M$, together with the explicit upper bound from Theorem 4.5.

Table 3: Example 1: numerical eigenvalue-localization summary for several M values.

M	$1/M$	# in-gap	$\max_{ E <g} \text{dist}(E, \text{spec}(A))$	bound
10.0	0.1000	4	0.1027	0.6591
20.0	0.0500	4	0.0817	0.2568
40.0	0.0250	5	0.0409	0.1156
50.0	0.0200	5	0.0328	0.0907
100.0	0.0100	5	0.0164	0.0437

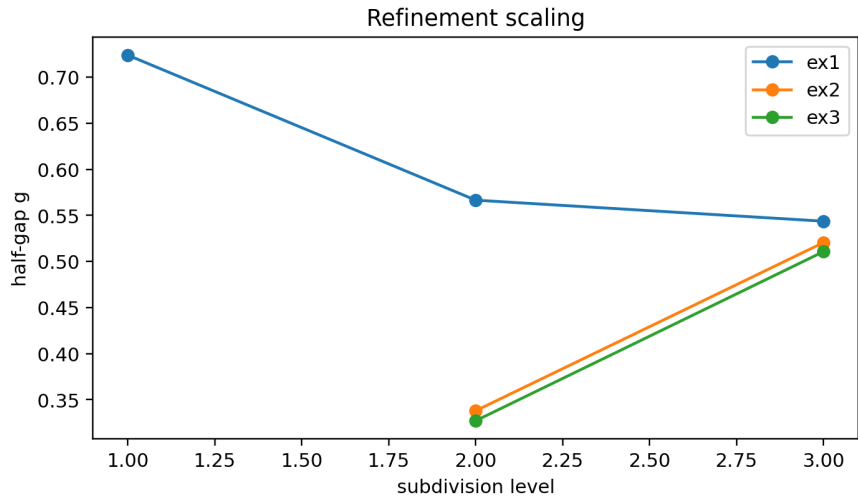


Figure 10: Bulk half-gap g versus subdivision level for the three example families.

7 Conclusion

The boundary case can be reduced to the closed universal model by a simple combinatorial operation: double the triangulated surface and jam the copy side. The unconditional mathematical content of this paper is finite-dimensional. We proved exact count identities for doubling, explicit $1/M$ decoupling estimates, a first-order effective operator, bounded-energy eigenvalue localization, leakage bounds, and a corrected low-energy spectral-subspace theorem giving convergence of the *full* spectral projections.

For bulk–interface physics, we made the external assumptions precise. Under a standard framework-admissibility hypothesis, doubling-and-jamming realizes a topological/trivial interface whose boundary class is the image of the bulk class of the doubled universal model under the standard boundary map. Thus the interface index is the Chern jump ± 1 , and its sign flips under $H_+ \leftrightarrow H_-$. The numerical experiments are consistent with this picture: the doubles are gapped at the tested resolutions, jamming creates in-gap states localized near the glued boundary, time evolution shows directed boundary transport, and the decoupling diagnostics follow the predicted $1/M$ and $1/M^2$ scalings.

A natural next step is to verify the framework-admissibility hypothesis directly for specific refinement families of triangulated surfaces, thereby placing the asymptotic bulk–interface statement fully inside a single coarse-geometric or interface-index formalism.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, numerical interpretations, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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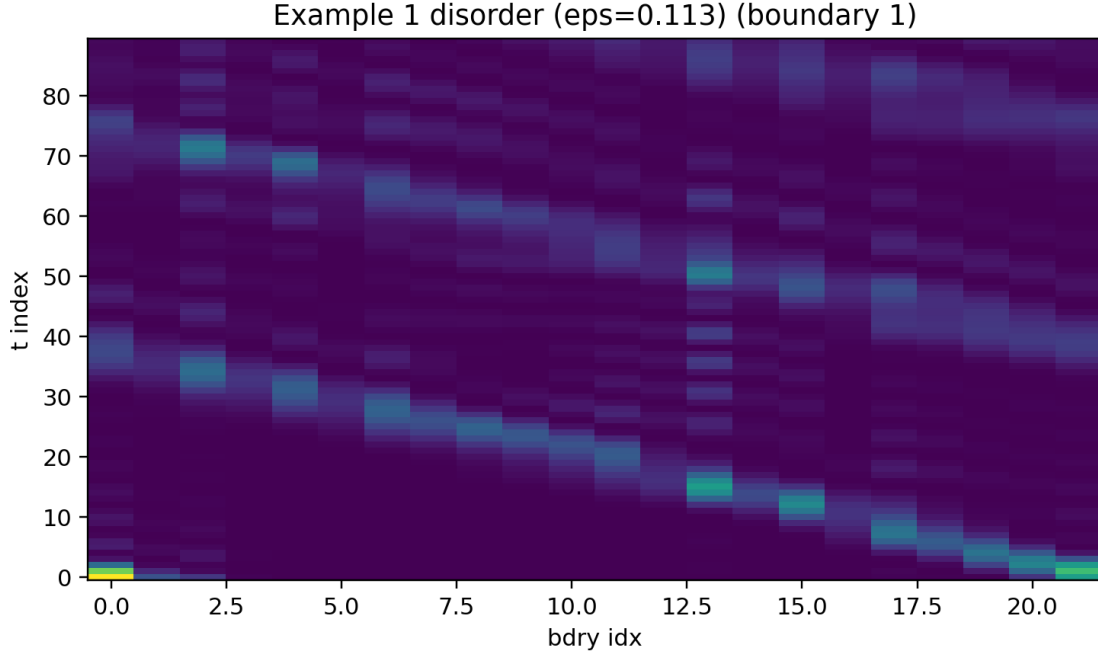


Figure 11: Example 1 with weak onsite disorder: the qualitative one-way boundary transport remains visible.

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